

Chapter 164 — Treatment of Locally Advanced Prostate Cancer

- **At least 10% of new diagnoses are locally advanced** (T3, NX or N+, M0) despite PSA-driven stage migration; the proportion has fallen slightly but has been broadly constant for a decade.
- **These men, with those presenting with metastases, account for a disproportionate share of prostate cancer deaths.**
- **No consensus on optimal management.** Unlike low-grade localised disease, where several modalities do equally well, **any single modality carries a significant risk of recurrence** → interest in combination therapy, better risk assessment and imaging.
- **Competing evidence:** several analyses favour radical prostatectomy (RP) for disease-specific mortality and metastasis in higher-risk men; other data favour external-beam radiation therapy (EBRT) + brachytherapy boost + androgen deprivation (AD).

DEFINITION

- **Traditional clinical definition** (digital rectal examination, DRE): extracapsular spread (**cT3a**), seminal vesicles (**cT3b**), adjacent organs (**cT4**).
- **PSA era:** most cancers are **non-palpable (cT1c)**, so PSA, Gleason and imaging, not T stage, now identify risk of failure.
- **Working definition:** "high-risk" disease, including **regional or nodal spread without distant metastasis** (T3–4, N0 or N+, M0).

Contemporary Risk Assessment

- **Combined variables beat single ones** for predicting pathological stage; many "organ-confined" tumours are more advanced. Knowing likely stage improves counselling, choice of therapy and trial selection.
- **Key pathological predictors after RP:** Gleason score, **margin status**, non-organ-confined disease (extracapsular extension [ECE], seminal vesicle invasion [SVI], nodes).
- **Biochemical disease-free survival (BDFS; PSA-free survival)** is the usual endpoint but **only an intermediate surrogate**; timing of recurrence and PSA kinetics refine risk.
- **Recurrence** reflects **unrecognised micrometastases** or **persistent locoregional disease**.
- **Low risk does well; intermediate and high risk are heterogeneous** and need finer nomograms.

RISK STRATIFICATION TOOLS COMPARED

Tool	Inputs	Output	Distinguishing feature or limit
Partin tables (1990s; updated 2001, 2013, 2017)	Clinical stage, PSA, Gleason	Predicted pathological stage at RP	Clinically localised disease before RP only
Kattan nomograms	Clinical stage, biopsy Gleason, pretreatment PSA; the RT versions add AD use, total dose and combination (EBRT + permanent brachytherapy)	Continuous risk of progression after local therapy	Most widely used. The RP nomogram was built on T1c–T3aNXM0 , so it does not apply with SVI or regional spread
D'Amico classification	Clinical stage, PSA, Gleason	Low / intermediate / high risk of biochemical failure	Simple. Variants add ethnicity and pathology

Tool	Inputs	Output	Distinguishing feature or limit
CAPRA score (Cancer of the Prostate Risk Assessment), 0–10	PSA, T stage, age , grade, cancer volume	Each 2-point rise doubles recurrence risk after RP	Validated in several populations; also predicts disease-specific and overall mortality
Clinical-genomic risk groups	Clinicopathological variables + 22-gene Decipher result	Three- or six-tier groups	Outperformed NCCN and CAPRA for metastasis and disease-specific survival; reclassified about two-thirds of NCCN six-tier patients

- **What the table tests:** which tool predicts *stage* (Partin) and which predicts *outcome* (Kattan, D'Amico, CAPRA, clinical-genomic).
- **Trap:** the Kattan RP nomogram **cannot be applied to cT3b or node-positive disease**, which is the population this chapter is about.

KEY POINTS: RISK ASSESSMENT

- **Risk assessment is best performed by combining** serum PSA, T stage, cancer grade and extent of cancer on biopsy.
- **Imaging has had a limited role** in identifying high-risk men in whom local therapy may fail, but newer modalities will change this.

Imaging Modalities

- **Grey-scale transrectal ultrasonography (TRUS)** is no better than DRE for ECE or SVI; early high accuracy was not reproduced. It **understages** because of interobserver variability, subtle signs and today's lower-volume tumours. **Directed biopsy of the seminal vesicles or capsule** can confirm cT3.
- **Colour/power Doppler and micro-ultrasound (29 MHz)** are under investigation.
- **Endorectal and multiparametric MRI (mpMRI)** may separate T2a/b from T3; its staging role and the need for the coil **remain controversial**. ECE accuracy varies widely (sensitivity 13–91%, specificity 49–97%) with interpretation, so **adopt PI-RADS v2 structured reporting**.
- **At 3 tesla, the PI-RADS v2 score next to the neurovascular bundle has poor specificity**, so it cannot by itself decide nerve preservation.
- **MRI is best reserved for higher-risk men** (more than three positive cores, abnormal DRE, PSA >10 ng/mL), where its specificity for pT3 is high.
- **Positron emission tomography (PET) tracers approved by the U.S. Food and Drug Administration:** ¹⁸F-fluciclovine for biochemical recurrence; ¹⁸F-sodium fluoride for bone.
- **Prostate-specific membrane antigen (PSMA) PET** detects limited nodal disease and oligometastases. For pelvic nodes, ⁶⁸Ga-PSMA-11 **specificity is very high (≥95%) but sensitivity only moderate (~40–66%)**. Whether it replaces standard imaging is debated.
- **proPSMA** (high-risk men; 30% had nodal or distant metastases): **PSMA PET-computed tomography (CT) accuracy 92% vs 65% for conventional imaging** (sensitivity 85%, specificity 98%), with lower radiation, excellent interobserver agreement, fewer equivocal findings and cost-effectiveness.

STAGING MODALITIES COMPARED

Modality	Strength	Weakness	Place
DRE	Defines cT3/cT4	Most cancers now cT1c	Baseline clinical stage

Modality	Strength	Weakness	Place
Grey-scale TRUS	Guides directed SV or capsular biopsy	Understages ; operator-dependent	No staging gain over DRE
mpMRI (± endorectal coil)	Zonal anatomy; T2 vs T3; high specificity for pT3 in high risk	Variable accuracy; poor specificity at the neurovascular bundle	Higher-risk men ; PI-RADS v2 reporting
Conventional imaging (CT/MRI + bone scan)	Standard, widely available	Less accurate than PSMA PET	High-risk staging (NCCN triggers below)
PSMA PET	Highest accuracy ; very high nodal specificity	Moderate nodal sensitivity	May substitute for CT + bone scan

Figure legends.

- **Fig. 164.1 (TRUS):** increased colour Doppler flow at the left posterior prostate, with a matching hypoechoic area. **Lateral distortion and "squaring" of the capsule suggest ECE.**
- **Fig. 164.2 (TRUS):** hypoechoic tumour at the left base **extending into the ipsilateral seminal vesicle.**
- **Fig. 164.3 (axial T2-weighted MRI):** tumour involving the seminal vesicle and **bladder base**, extending into the adjacent fat.
- **Fig. 164.4 (⁶⁸Ga-PSMA-11 PET/MRI):** uptake in the right peripheral zone and in **right pelvic side-wall and external iliac nodes.**

Novel Markers

- **Clinical and pathological parameters predict local extent poorly.** Comparative genomic hybridisation aberrations discriminate organ-confined from locally advanced tumours (Table 164.1); a stepwise six-aberration model was highly accurate.
- **TMPRSS2-ETS rearrangements** (androgen-regulated) → **higher pathological stage**; **sarcosine** (glycine metabolite) raised in metastatic disease.
- **CALGB 90203 profiling** (high-risk localised): frequent **TMPRSS2-ERG fusion, TP53, PTEN, FOXA1, SPOP** alterations but **no androgen receptor (AR) amplification or mutation**; marked genomic and immune-infiltrate heterogeneity; Wnt signalling correlated with immune suppression.
- **Commercial genomic classifiers** (Decipher, Oncotype DX, Prolaris) are applied variably to refine risk (NCCN 2023).
- **After RP with adverse pathology**, genomics identifies higher disease-specific mortality and may guide adjuvant therapy. **Unresolved:** how to use these tests and whether they improve outcomes.

TABLE 164.1 CHROMOSOME ABNORMALITIES ASSOCIATED WITH PATHOLOGICAL STAGE

Chromosome	Aberrations associated with stage (alone or in combination; – loss, + gain)
8p	–8p alone; –8p with –10q23→qter, –5q31→qter, –15q22 or –11q24→qter
10q	–10q25→qter; –10q24→qter with –5q31, –6p21 or +7p11
6	–6p21; –6q24→qter; –6p12→pter with –6q24→qter; –6q25→qter with –11q23→qter
18	–18cen–q12

- **Pattern:** almost all are **losses**, clustered on **8p, 10q and chromosome 6**. The single gain is +7p11, and it occurs only in combination with a 10q loss.
- **Purpose:** it shows that a genomic signature, not just clinical variables, can predict extraprostatic disease.

Concept — *Locally advanced disease is now defined by risk, not palpation. Because clinical staging and imaging both understage, every tool in this section is an attempt to predict the pathology that surgery would reveal and the failure that monotherapy would produce.*

TRENDS IN INCIDENCE AND TREATMENT

- **CaPSURE (Cancer of the Prostate Strategic Urologic Research Endeavor):** D'Amico high-risk presentation fell from about 41% (1989–1990) to 15% (2001–2002), driven by PSA screening (far fewer PSA >20 ng/mL); clinical T3–T4 fell from about 12% to 3.5%. **High Gleason grade now defines most high-risk patients**, as also seen at Johns Hopkins.
- **National Cancer Database (2004–2016):** high-risk proportion **rose** (about 12% → 20%) with RP use (about 23% → 41%).
- **cT3 and pT3 at RP have fallen steeply** (cT3 about 25% in 1987 → under 3% in 2001) through PSA screening, selective surgery and better RT; ECE fell across every clinical, PSA and Gleason stratum (Table 164.2).
- **NCCN (National Comprehensive Cancer Network) 2023:** high risk includes **cT3a and cT3b–T4. Bone scan + CT or MRI** for these and for **PSA >20 ng/mL, grade group 4–5 or primary Gleason pattern 5; PSMA PET may substitute** (more sensitive).
- **RP + extended pelvic lymphadenectomy only for high-risk tumours that can be completely excised;** the alternative is **AD + RT**.
- **The pattern has swung back:** historically low RP rates (CaPSURE; Surveillance, Epidemiology, and End Results [SEER]) have given way to a **significant rise in surgery for CAPRA 6–10**.

TABLE 164.2 PATHOLOGICAL FINDINGS IN MALES UNDERGOING RADICAL PROSTATECTOMY

Series grouped by era	ECE	SVI	Lymph node positive
Series spanning the 1980s–1990s	25–66%	4–18%	2–7%
Same-institution comparisons, earlier → later era	60% → 25%; 66% → 25%	18% → 5%	SV + LN combined 14% → 4%
Series from 2000 onwards	~14–17%	~5–6%	up to 9%
University of California, San Francisco (UCSF), 1986–2009	16%	11%	3%

- **Pattern:** ECE and SVI at RP have **fallen severalfold** across eras within the same institutions. This reflects stage migration and selection, not any change in biology.
- **Trap:** a falling pT3 rate at RP does not mean locally advanced disease has disappeared. Surgical series are **selected** for largely clinically localised disease.
- **Fig. 164.5 (CaPSURE, CAPRA 6–10, 1990–2013; Cooperberg and Carroll, 2015):** primary AD peaked in 2005–2009 then fell; RP became commonest by 2010–2013; RT declined; surveillance stayed minimal.

KEY POINTS: TRENDS IN INCIDENCE AND TREATMENT

- **Fewer men present** with locally advanced prostate cancer.
- **Organ-confined cancers found at RP have increased.**
- **Non-surgical treatment of high-risk cancer increased**, but this has changed in recent years.

Concept — Stage migration has redefined "high risk". What used to mean a palpable T3 tumour now mostly means high Gleason grade. Contemporary high-risk cohorts are therefore less advanced than the historical series on which treatment was built, and that is part of why surgery has returned.

NATURAL HISTORY

- **Active surveillance with deferred treatment is viable**; disease-specific mortality is low and concentrated in higher grade and stage, though even low-risk cancers can progress over long follow-up. Genomic tests help selection. **Higher-risk disease progresses faster**, so healthy men usually warrant intervention.
- **Few relevant series of observed locally advanced disease**; older ones are of little relevance, and reported 5–10-year progression, metastasis and survival vary widely.
- **Veterans Administration Cooperative Urological Research Group (VACURG) placebo arm, stage III**: 5-year OS 58%. **Medical Research Council (MRC) delayed-AD arm** (locally advanced M0): median time to clinical progression **10 months**, to prostate cancer death **48 months**.
- **Grade drives cancer death under conservative management**: grade 3 carries about **tenfold** the disease-specific mortality of grade 1.
- **Swedish population cohort** (Table 164.3): in localised disease, deferred and initial treatment gave similar 15-year corrected survival; locally advanced disease did worse. Cancer death rose from rare (well differentiated) to over half (poorly differentiated).
- **Connecticut Tumour Registry** (pre-PSA watchful waiting): 15-year cancer mortality rises with Gleason (6 < 7 < 8–10; **Gleason 8–10: 60–87%**); in Gleason 6, competing mortality exceeds it. **Annual mortality from low-grade cancer is stable beyond 15 years**, unlike the late rise in the Swedish cohort.
- **SEER contemporary cohort**: better outcomes, but **Gleason 8–10 still >20% cancer-specific mortality at 15 years**. **Untreated high-risk disease carries significant risk of progression and cancer death**.
- **Undertreatment**: in Sweden, **fit men in their 70s** with life expectancy >10 years received RT in only 40% and surgery in 10%, despite life expectancy comparable to younger treated men; the UK National Prostate Cancer Audit found the same.

TABLE 164.3 CONSERVATIVE MANAGEMENT OF PATIENTS WITH PROSTATE CANCER

Clinical stage	Progression-free survival	Disease-specific survival
T1–T2	48%	81%
T3–T4	47%	57%
M+	6%	6%

From Johansson et al., 1997 (JAMA): 15-year survival, prospective population-based study in Sweden.

- **Pattern**: T1–T2 and T3–T4 **progress at the same rate** but **die of cancer at very different rates**. Locally advanced disease converts progression into death. Metastatic disease is almost uniformly fatal.
- **Trap**: equal progression-free survival does not imply equal disease-specific survival.
- **Fig. 164.6 (Albertsen et al., 2005; ages 55–74)**: **Gleason 8–10 — prostate cancer death dominates at every age**; Gleason 7 — competing mortality grows with age.

Concept — Grade drives cancer death in untreated disease, and competing mortality decides whether that matters for a given man. A fit man in his 70s with high-grade disease will live long enough to die of it, which is why withholding treatment from him is an error.

RADICAL PROSTATECTOMY

- **RP for locally advanced disease declined**: surgery alone is often insufficient, risk assessment improved, RT delivery improved and **combined RT + AD beat monotherapy**. Yet **RP can cure some high-risk men**, and adding secondary therapy may improve on it.

Surgery for Clinical Stage T3 Prostate Cancer

- **OS after RP for cT3** (all series): **64–96% at 5, 12.5–72% at 10, 20–51% at 15 years**; older series carried inaccurate staging and occult nodes.
- **Contemporary disease-specific survival: 85–92% at 5 years and 79–82% at 10 years, regardless of adjuvant therapy.**
- **Surgery alone in cT3a:** about half recurrence-free at 8–10 years without adjuvant AD. **Mayo Clinic cT3** at 20 years: local recurrence-free, systemic progression-free and disease-specific survival 72–81%.
- **Clinical overstaging of cT3 occurs in 7–27%, so men should not be excluded from RP on clinical stage alone.**
- **Higher survival in men who completed RP with node dissection; over a third of high-risk RP specimens are specimen-confined (pT2–pT3a),** predictable by a nomogram (PSA, age, clinical stage, biopsy Gleason).
- **Selection is key:** many cT3 tumours have regional spread, but in **lower-volume disease** local control is usual and complete excision possible.
- **Biochemical progression** is confounded by adjuvant therapy: **>60% at 5 years without secondary treatment;** 42–49% (5 years) and 59–62% (10 years) with variable adjuvant use.
- **Clinical progression** (biopsy-proven local or objective metastasis): **12–45%, 39–49%, 50–71% at 5, 10, 15 years,** little affected by adjuvant therapy. **Survival benefit over alternatives is unproven.**

TABLE 164.4 RADICAL PROSTATECTOMY FOR HIGHER CLINICAL STAGE IN CONTEMPORARY SERIES

Series grouped by era	Adjuvant therapy	OS 5 yr	OS 10 yr (15 yr)	DSS 5 yr	DSS 10 yr (15 yr)
1991–1999 series	0–60%	75–86%	60–72%	85–90%	70–82%
2005–2009 series	0–62% (one neoadjuvant)	85–91%	74–76% (53–69%)	90–98%	85–91% (76–84%)
Surgery alone, no adjuvant (Johns Hopkins)	0%	—	—	BDFS 60%*	BDFS 49%*

*BDFS defined as PSA ≤0.2 ng/mL; 15-year actuarial metastasis-free survival 82% in the whole cohort (updated data from Han et al., 2001). One series reports 7-year figures. OS, overall survival; DSS, disease-specific survival.

- **Pattern:** DSS stays high (80–90% at 10 years) even where BDFS is only about half. **Biochemical failure after RP for cT3 is common, but cancer death is not.**
- **Values that differ:** the body text gives contemporary DSS as 85–92% (5 years) and 79–82% (10 years). The table's 2005–2009 group reads higher (90–98%, 85–91%). The difference is grouping: the text's "contemporary" range includes 1990s series.
- **Trap:** do not confuse the wide OS ranges (dominated by old, understaged cohorts) with contemporary DSS.

Outcomes of Prostatectomy for Pathologically Advanced Disease

- **A significant minority of clinically organ-confined tumours are pT3. cT3 tumours** tend to be larger, higher grade and more often regionally spread than upstaged pT3; most pT3 today are high risk by PSA or Gleason, and the groups overlap. Pathological stage is a powerful predictor (Fig. 164.7).
- **Positive margin carries the greatest relative risk** — more than ECE, SVI or nodes — so **completely remove all prostatic tissue regardless of stage.**
- **SVI, node-negative:** 15-year DSS about 80% although BDFS only about a third.
- **Biochemical recurrence depends on threshold:** 0.2 ng/mL → up to **65–72% at 5 years**; 0.4 ng/mL → about 26% at 5 and ≥50% at 10 years.

- **pT3bN0 at 10 years:** biochemical recurrence-free about 25%, metastasis-free 70%, DSS 80%.
- **SVI also raises local recurrence** (historically nearly half), now lower through selection, technique and earlier secondary treatment.
- **Minimally invasive RP in high risk: oncological and functional outcomes comparable to open;** lymphadenectomy extent varies.
- **PSA-free survival after RP in high-risk men $\approx$ 50% at 5–7 years;** higher biopsy Gleason and PSA raise the risk.
- **Kattan preoperative nomogram, community cohort** (concordance index 0.68): **underestimated recurrence in lower-risk and overestimated it in higher-risk men.**
- **Up to half with Gleason 8–10 or PSA >20 ng/mL have prolonged disease-free survival after RP alone;** better markers must identify them.

PATHOLOGICAL STAGE AFTER RP AND OUTCOME

Pathological finding	5-yr clinical progression	5-yr PSA-free survival	Relative risk of progression
Organ-confined	7%	95%	reference
ECE (focal / established)	18% / 35%	76%	2.17–2.72
SVI	86% (overall)	37%	2.61
Lymph node positive	95% (overall)	18%	3.31
Positive surgical margin	—	—	4.37 (highest)

- **Fig. 164.7 (Bianco, Scardino and Eastham, 2005):** cancer control after RP **plateaus near 90% if organ-confined**, about two-thirds with ECE; **SVI and N+ fall below 40% by about 10 years**, N+ lowest.

Outcomes of Prostatectomy for Node-Positive Disease

- **Renewed interest in surgery for N+ disease** (multimodal therapy, longer life expectancy, oncological and symptomatic benefit of removing the primary). Selection is unresolved; ^{68}Ga -PSMA-11 PET may identify limited or excisable disease.
- **Feasibility: cytoreductive RP for M1** has acceptable complications and continence; **surgery after neoadjuvant systemic therapy does not appear to raise perioperative complications** (CALGB 90203 to address).
- **N+ after RP:** about half alive at 10 years. **Single positive node:** nearly half without clinical progression, most never needing AD, DSS >80%.
- **N+ without adjuvant AD or RT** can still achieve reasonable 10-year OS and DSS. **Poor predictors: Gleason >8 and >3 positive nodes.**
- **Oligometastatic (≤ 5 bone metastases),** RP + extended node dissection + almost universal adjuvant AD (small cohort): about half progression-free, most free of cancer death at about 5 years.
- **Nodal disease as first site of clinical recurrence** carried better DSS. Oligometastatic local-therapy trials are ongoing.

Neoadjuvant Androgen Deprivation

- **History:** 1944, radical perineal prostatectomy after orchiectomy; later, diethylstilbestrol (DES) before surgery; a 1964 series had about half alive and recurrence-free at 10 years. Randomised cT1–T3 trials: Table 164.5.
- **Endpoints studied** included DRE, imaging, circulating cells and pathology, but **what matters is DSS or its surrogate BDFS.**

- **The biological effect is real:** PSA falls about 90%, **prostate volume 30–50%**, with spectroscopic and histological atrophy (fibrosis, vacuolisation, glandular collapse), **maximal within 2 months**.

CLINICAL VERSUS PATHOLOGICAL DOWNSTAGING AFTER NAD

Clinical downstaging	Pathological downstaging
Frequent: 32–90%	Uncommon: 8–31%. Only about 20% of cT3 tumours are organ-confined at RP after NAD
• Randomised cT2–T3 trial: 3 months of total androgen blockade doubled pathological downstaging overall, but not in cT3 (nor nodal metastases). • SVI and nodal rates are unchanged (Table 164.6). • Long-term (median 6 years), 3 months NAD vs surgery alone: no overall biochemical benefit; subgroup PSA >20 ng/mL did better (53% vs 35%). • Southwest Oncology Group (SWOG) 9109 (phase II, T3–T4N0M0; goserelin + flutamide 16 weeks): feasible, acceptable morbidity, outcomes comparable to RT (5-year progression-free 70%, OS 90%; at 10.6 years 40% and 68%). • Margins: NAD does not significantly reduce positive margins in cT3; in cT1–T2 it clearly does, but without long-term PSA-free gain. For cT3, no significant benefit of NAD before surgery (Table 164.7). • Too short? Combined blockade for more than 3 months lowered PSA failure in one cohort (hazard ratio [HR] 0.52). • Canadian Urologic Oncology Group, 3 vs 8 months NAD before RP: 8 months → positive margins 23% → 12%, organ-confined 68% → 80%, yet no difference in PSA recurrence at 4 years. • Newer agents (degarelix, abiraterone, enzalutamide): abiraterone + leuprolide + enzalutamide → complete response or minimal residual disease about 30%; enzalutamide alone, no pathological complete response; enzalutamide + leuprolide + dutasteride about 17%.	

TABLE 164.5 PROSPECTIVE RANDOMIZED TRIALS OF NEOADJUVANT ANDROGEN DEPRIVATION IN PATIENTS WITH PROSTATE CANCER (STAGE CT1–T3) BEFORE RADICAL PROSTATECTOMY

LHRH agonist	Antiandrogen	Duration (months)	Trials
Leuprolide	Flutamide	3	2
Goserelin	Flutamide	3	1
Triptorelin	Cyproterone	3	1
Goserelin	—	3	1
—	Cyproterone	3	1
Estramustine (antimicrotubule agent derived from oestradiol; also lowers testosterone)		1.5	1

LHRH, luteinising hormone–releasing hormone.

- **Pattern:** almost every trial used **3 months** of AD, which is the basis of the "too short" critique. The one outlier used estramustine for 1.5 months.

TABLE 164.6 PATHOLOGICAL FINDINGS FROM STUDIES OF NEOADJUVANT ANDROGEN DEPRIVATION IN PATIENTS WITH CT1–T3 PROSTATE CANCER BEFORE RADICAL PROSTATECTOMY

Finding	Control (surgery alone)	NAD
Seminal vesicle invasion	6–23%	0–28%
Lymph node metastasis	5.8–23%	3.3–15%

- **Pattern:** the ranges overlap almost completely. NAD does not clear the seminal vesicles or nodes.
- **Trap:** one trial found significantly **higher** SVI after NAD (28% vs 14%). NAD does not reduce SVI.

TABLE 164.7 NEOADJUVANT ANDROGEN DEPRIVATION THERAPY FOR 3 MONTHS BEFORE RADICAL PROSTATECTOMY IN PATIENTS WITH LOCALIZED PROSTATE CANCER: IMPACT ON PSA RECURRENCE AFTER SURGERY

Arm	PSA recurrence	Follow-up
Control	16–41%	29–60 months
NAD (3 months)	11–40%	

- **Pattern:** there is no consistent direction. In several trials PSA recurrence was numerically higher with NAD. Three months of NAD does not reduce biochemical failure.

Neoadjuvant Chemotherapy and Chemotherapy–Hormonal Therapy

- **Chemotherapy was confined to the most advanced disease:** taxanes in hormone-refractory disease give PSA falls >50% in over half and responses in 28–75%; **mitoxantrone + low-dose steroid** improves pain relief and is approved. Hence earlier use in high risk (Table 164.8).
- **CALGB 90203** (neoadjuvant AD + docetaxel before RP): **primary endpoint (3-year biochemical recurrence) not met**, but overall biochemical recurrence, **MFS and OS improved**.
- **Estramustine + etoposide without NAD:** grade 3–4 toxicity in about a third, surgery never delayed, outcomes comparable; early failures were node-positive.
- **Docetaxel alone and nanoparticle paclitaxel:** well tolerated, **no pathological complete responses**; docetaxel gave PSA falls >50% in a proportion.
- **Docetaxel + dose-escalated mitoxantrone** (16 weeks): half relapse-free at 5 years; margins positive in a third.
- **Also tried:** granulocyte–macrophage colony-stimulating factor + thalidomide; gefitinib (epidermal growth factor receptor inhibitor). Roles of **custirsén** and **immunotherapy (sipuleucel-T, ipilimumab, PD-1/PD-L1 inhibitors)** undetermined.

TABLE 164.8 TRIALS OF CHEMOTHERAPY, HORMONAL THERAPY, AND COMBINED CHEMOTHERAPY–HORMONAL THERAPY BEFORE RADICAL PROSTATECTOMY

Regimen	Agents	With AD	Organ-confined	Positive margins	LN+	pT0
KAVE	Ketoconazole + doxorubicin/vinblastine + estramustine	3 mo	33%	17%	37%	0%
E/VP-16	Estramustine + etoposide	No	31%	44%	13%*	0%
TEC	Paclitaxel + estramustine + carboplatin	4–6 mo	36%	22%	5.5%	0%
D	Docetaxel alone	No	18%	25%	14%	0%
D + AD	Docetaxel	6 mo	53%	27%	6%	3%
NAD alone (23 cT3 trials pooled)	—	Yes	21%	37%	21%	0–4%

*Grouped with seminal vesicle invasion. The proportion of cT3 disease varied from 13% to 75% between regimens (100% in the pooled NAD trials). SVI after chemotherapy regimens was 22–85%. Undetectable preoperative PSA occurred in about half of men on KAVE, E/VP-16 and TEC.

- **Purpose:** to compare chemotherapy-containing regimens against NAD alone in cT3 disease.
- **Pattern:** **pT0 is essentially never achieved (0–4%).** Docetaxel with 6 months of AD gave the most organ-confined disease.
- **Trap:** an undetectable preoperative PSA after neoadjuvant therapy does not mean pathological complete response.

Adjuvant Radiation Therapy

- **Why defer further therapy until after RP:** no delay to surgery, less operative morbidity, and **most importantly, pathology identifies who truly needs it**, sparing the rest. The difficulty: benefit depends on the **site of failure** (local or distant), which is unknown.
- **Older studies: no effect on metastasis or DSS; newer data: better biochemical control.** Adjuvant RT BDFS 50–88% at 5 years vs 30–50% after surgery alone in high risk (Table 164.9).
- **Immediate RT vs observation** (European Organisation for Research and Treatment of Cancer [EORTC] 22911, SWOG 8794, ARO 96-02; table below) **reduced clinical local recurrence**.
- **EORTC 22911 detail:** the positive-margin benefit (HR 0.38) held regardless of SVI or a postoperative PSA above or below 0.2 ng/mL; **even pT3b with a positive margin benefited**.
- **SWOG 8794 detail** (median 10.6 years; PSA <0.4 ng/mL after RP): biochemical recurrence **35% vs 64%**; metastases **16% → 7%**; less subsequent AD. **With SVI**, 10-year BDFS improved (36% vs 12%) but **MFS and OS did not**.
- **ARO 96-02 detail:** ECE without SVI predicted a more favourable response to RT.
- **Adverse features are not equivalent**, yet SWOG 8794 pooled ECE, SVI and margins.
- **Decision analysis:** benefit is confined to likely **local** recurrence (low–intermediate grade, no SVI); **consider adjuvant RT for extensive or multiple positive margins**.
- **Salvage RT:** about half progress afterwards; high Gleason and SVI predict poor response, yet selected adverse-feature men (high grade, rapid PSA doubling time) respond durably — over half with **Gleason 8–10 and positive margins** reached a nadir ≤0.1 ng/mL without rise.
- **Early secondary therapy** (RT or AD) may help selected men; salvage RT in pT3 was associated with better OS. Optimal timing and effects on local control, metastasis and survival remain undefined.
- **pT3b:** high distant risk, but some recur only locally; with undetectable PSA, adjuvant RT improved 3-year BDFS.
- **Favourable pT3b for adjuvant RT:** PSA <0.3 ng/mL after RP or **positive margins**. A PSA never reaching undetectable → occult nodal or distant disease, poor prognosis.
- **24-gene signature:** high-score men given postoperative RT had less 10-year metastasis, so it may predict RT response.
- **Adjuvant versus early salvage RT** (RADICALS-RT, GETUG-AFU 17, RAVES and a pre-planned meta-analysis): **early salvage is preferred for most men**.
- **Trap:** very high-risk men (**Gleason 8–10, pT3b**) made up only ~20% of these trials, so adjuvant RT may still benefit them.
- **Dose:** traditionally lower than salvage (45 to >60 Gy); contemporary >60 Gy. BDFS better **above 61.2 Gy**; with detectable PSA, **≥64 Gy** improved freedom from failure (a dose response like primary RT). **RAVES, RADICALS-RT, GETUG-AFU 17 used 64–66 Gy**.
- **RTOG 0621** (Radiation Therapy Oncology Group; high risk after RP, mostly pT3–T4, Gleason ≥8): 6 months AD + RT + six cycles docetaxel → **3-year freedom from progression 73% vs 50% expected**, supporting phase III trials.
- **GETUG-AFU 16:** 3 months of AD with RT suffices — 10-year PFS 64% vs 49% (HR 0.54).

TABLE 164.9 EARLY ADJUVANT RADIATION THERAPY FOR LOCALLY ADVANCED PROSTATE CANCER (PT3N0M0)

Arm	Progression-free survival	Radiation dose	Follow-up
Observation	37–69%	—	11–151 months
Adjuvant RT	55–94%	48–70 Gy	

Arm	Progression-free survival	Radiation dose	Follow-up
SWOG 8794 (observation vs RT)	10-year metastasis-free survival 61% vs 71% (60–64 Gy; median 151 months)		
Split by margin (one series)	Negative/positive margin: 53%/37% observed vs 62%/38% with RT (not significant)		

Gy, gray.

- **Pattern:** adjuvant RT is better in most series, but the **non-randomised comparisons were mostly small** and several were not significant.
- **Trap:** the only survival-level evidence in the table is the randomised SWOG 8794 row.

POSTOPERATIVE RADIATION THERAPY TRIALS COMPARED

Trial	Population	Comparison	Primary endpoint	Verdict
EORTC 22911	pT3 and/or margin +, N0	Immediate RT vs observation	BDFS	BDFS 74% vs 53%; benefit only with positive margins
SWOG 8794	pT3 and/or margin +, N0	Immediate RT vs observation	MFS	MFS (HR 0.71) and OS (HR 0.72) improved
ARO 96-02	pT3N0, undetectable PSA	Adjuvant RT vs observation	BDFS	BDFS HR 0.53
RADICALS-RT	Post-RP	Adjuvant vs early salvage	Freedom from distant metastasis	Early salvage preferred; 5-yr event-free 87–94% in both arms
GETUG-AFU 17	Post-RP	Adjuvant vs early salvage	Event-free survival	
RAVES	Post-RP	Adjuvant vs early salvage	Biochemical recurrence (non-inferiority)	

KEY POINTS: ADJUVANT RADIATION THERAPY

- **Observation with early salvage RT** appears preferable after prostatectomy.
- **Benefit of adjuvant RT may be limited to very high-risk disease.**
- **Dose escalation (≥64 Gy)** is associated with better adjuvant RT outcomes.

Concept — Adjuvant RT treats a local problem. *Its benefit concentrates where residual disease is local (a positive margin) and fades where failure is systemic (SVI, a PSA that never becomes undetectable). That is why, for most men, observation with early salvage matches immediate treatment.*

Adjuvant Androgen Deprivation

- **Rationale:** early-AD benefit in MRC and VACURG, and continued AD after RT.
- **pT3N0 disease: no convincing benefit.** Adjuvant (mostly endocrine) therapy did not improve time to progression or survival. At the Mayo Clinic (pathological stage C), adjuvant AD or RT reduced progression **but not DSS or OS**, with no difference between the two. Adjuvant goserelin improved disease-free survival.
- **pT3b and limited N+: benefit** — early adjuvant AD improved time to progression and DSS; **single positive node** → 10-year cause-specific survival 94%.
- **ECOG 7887** (Eastern Cooperative Oncology Group; cT1–T2 with nodes at RP; median 11.9 years): the HRs for overall, disease-specific and progression-free survival (1.84, 4.09, 3.42) are **expressed for delayed relative to immediate AD**.
- **pN1: adjuvant AD + RT beats AD alone.** In a matched-pair analysis, 10-year DSS was 86% vs 70% and OS 74% vs 55%.

- **Observation vs lifelong AD vs AD + RT in node-positive men:** AD + RT OS HR 0.41–0.46, with better DSS and more benefit in higher risk. **AD alone raised other-cause mortality (HR 3.05).**

ADJUVANT SYSTEMIC THERAPY AFTER RP COMPARED

Trial or series	Population	Intervention	Verdict
ECOG 7887	pN+ after RP	Immediate vs delayed AD	OS, DSS and PFS improved
Node-positive cohort	pN+	Observation vs AD vs AD + RT	AD + RT best OS; AD alone ↑ other-cause mortality
Retrospective series	pT3N0	Adjuvant AD	Less progression; no DSS or OS gain
SWOG S9921	High risk after RP	AD ± mitoxantrone	No benefit
VA CSP #553; SPCG-12	High risk after RP	Docetaxel	No benefit

KEY POINTS: RADICAL PROSTATECTOMY

- **RP alone gives disease-free survival in at least half of men at 8–10 years**, despite clinically advanced disease.
- **Recurrence risk after surgery is quantified by pathological stage, cancer grade and surgical margin status.**
- **NAD before RP does not improve disease-specific or overall survival.**

Concept — Adjuvant AD after RP helps only once disease has left the prostate (N+, pT3b), not in pT3N0. In node-positive disease the survival gain comes from AD combined with local RT. AD alone carries its own non-cancer mortality.

RADIATION THERAPY

- **The trend is non-surgical treatment of high-risk disease.** Many older cT3 series predate PSA and modern techniques — three-dimensional conformal RT (3D-CRT), intensity-modulated RT (IMRT), whole-pelvis RT.
- **Dose: ≥81 Gy** improves biochemical control in higher risk; **78 vs 70 Gy** helped most with **PSA >10 ng/mL**.
- **EBRT or permanent brachytherapy monotherapy is probably inadequate;** the focus is RT + AD, often with whole-pelvis RT (Table 164.10).
- **Trap: the demonstrated benefits of AD + RT over RT alone often do not include overall survival.**
- **RT alone, stage C:** OS 60–70% at 5 years, <50% at 10; high-risk 5-year PFS typically <50%.

Neoadjuvant Androgen Deprivation and Radiation Therapy

- **Rationale:** smaller target volume and cytotoxic synergy with radiation. Early RTOG trials (75-06, 83-07, 85-19): tolerable, better local control.
- **RTOG 86-10** (bulky cT2–T4): goserelin + flutamide for 2 months before and 2 during RT (**4 months**) reduced distant metastases and **disease-specific mortality (23% vs 36% at 10 years)**. **OS did not differ** (Table 164.10); only Gleason 2–6 appeared to gain survival, and Gleason 7–10 gained neither locoregional control nor survival.
- **RTOG 86-10 correlatives:** loss of p16 → worse OS, DSS, local progression and metastasis (not OS on multivariate analysis); high Ki-67 → metastasis and DSS, not OS.
- **Canadian three-arm trial** (RT vs 3 months NAD + RT vs NAD + RT + 6 months adjuvant AD): **residual cancer on biopsy at 24 months 65% → 28% → 5%**, BDFS in parallel — supporting neoadjuvant, concurrent and prolonged AD.

- **3 vs 8 months of NAD before RT:** no change in freedom from or pattern of failure; only 5-year DFS improved (see the note after Table 164.10).
- **TROG 96.01** (Trans-Tasman Radiation Oncology Group; 66 Gy; median 10.6 years): 6 months of NAD reduced PSA and local progression, improved event-free survival (HR 0.51), and reduced distant progression, disease-specific and all-cause mortality (HR 0.49, 0.49, 0.63); **3 months did not affect the last three.**
- **RTOG 9910** (8 vs 28 weeks NAD, each + 8 weeks concurrent): **no general benefit in prolonging NAD;** applicability to high risk unknown.

NEOADJUVANT AD BEFORE RP VERSUS BEFORE RT — THE GOVERNING ASYMMETRY

NAD before RP	NAD (+ concurrent AD) with RT
Shrinks the gland and clinically downstages, but rarely downstages pathologically	Shrinks the target volume and is synergistic with radiation
No gain in BDFS, DSS or OS	Improves local control, DFS, metastasis and disease-specific mortality (RTOG 86-10, TROG 96.01)
Longer course (8 months) improves pathology but not PSA recurrence	6 months better than 3 months (TROG 96.01); for high risk, prolonged adjuvant AD is better still

KEY POINTS: ANDROGEN DEPRIVATION AND RADIATION THERAPY

- Neoadjuvant and concurrent AD appears appropriate in high-risk patients undergoing RT.
- Adjuvant AD after RT may benefit those with very high-risk disease.

Adjuvant Androgen Deprivation and Radiation Therapy

- **RTOG 85-31** (under 30% N+): goserelin from the last week of RT **indefinitely** vs at relapse. Better local control, BDFS and MFS, **most in Gleason 8–10; OS not different overall** (5 years 75% vs 72%); in high grade, 5-year DSS 90% vs 78% and **OS 80% vs 69%**.
- **EORTC 22863** (goserelin from the start of RT for **3 years**): **the only study in this set with an OS benefit** (5-year OS 79% vs 62%); local control 97% vs 79%. The cohort was very high risk (median PSA 30 ng/mL), which may explain poor survival with RT alone.
- **The method of AD does not matter.** In the Swedish orchiectomy + RT trial, DSM did not differ and **node-negative men had no survival difference**; poor survival came mainly from metastases present at treatment.
- **RTOG meta-analysis** (trials 1975–1992): **risk group 2** (cT3NX Gleason 2–6; cT1–T2NX Gleason 7; N+ Gleason 2–6) gained DSS from **4 months of AD**; bulky or cT3 disease gained survival.
- **RTOG 92-02** (4 months AD, 2 neoadjuvant + 2 concurrent, vs + **24 months**): better on **every endpoint except OS**; **Gleason 8–10 OS 81% vs 71%**, confirmed at 10 years.
- **70 Gy ± 6 months AD** (PSA >10, Gleason ≥7 or cT3): **OS and DSS improved with AD.**
- **With high-dose RT, 2 years of AD beat short-term concurrent AD** for biochemical recurrence, OS and MFS. **18 vs 36 months** (prostate + pelvic RT): OS comparable; 18 months better on several quality-of-life scales.
- **Bottom line:** intermediate risk → **2–4 months**; high risk → **prolonged AD**. NCCN: **18–36 months with EBRT; 12–36 months with brachytherapy boost.**
- **Clinical cell-cycle risk score** (CAPRA + cell-cycle progression molecular score): identifies a group in whom adding AD to RT did not reduce 10-year metastasis.

- **RTOG 94-13** (nodal risk >15%; 2 × 2: whole-pelvis vs prostate-only RT; neoadjuvant + concurrent vs adjuvant AD for 4 months): early **PFS advantage for whole-pelvis** disappeared; **OS unexpectedly worst with whole-pelvis + adjuvant AD**.
- **Dose >72 Gy** is more effective, and differences in dose make cross-trial comparison difficult.
- **Brachytherapy in high risk** is less studied: permanent or high-dose-rate + EBRT gave moderate BDFS; permanent + 45 Gy EBRT **without AD**, about half at 10 years. **Trimodality** (9 months AD + brachytherapy + 3D-CRT): 5-year BDFS 86%, Gleason the only predictor, no cancer on final biopsy.
- **ASCENDE-RT** (NCCN intermediate and high risk; 12 months AD + pelvic RT 46 Gy + boost): **EBRT boost 78 Gy vs low-dose-rate (LDR) brachytherapy boost — OS equal; biochemical recurrence doubled with EBRT boost (HR 2.04)**; 9-year 85% vs 65%, diverging after 4 years.
- **Gleason 9–10 (retrospective): EBRT + brachytherapy boost** lowered cancer-specific mortality (HR 0.38–0.41) and metastasis (HR 0.27–0.3) vs RP or EBRT alone, **despite shorter AD** (12 vs 22 months).

TABLE 164.10 TRIALS OF ANDROGEN DEPRIVATION WITH EXTERNAL-BEAM RADIATION THERAPY

Trial	Eligibility	RT dose (Gy)	Arms	DFS	OS	Median follow-up
NEOADJUVANT						
RTOG 86-10	cT2–4, ≥25 cm ²	44–46 WP; 65–70 P	RT vs RT + nAD + cAD	3% vs 11%*	34% vs 43%†	12.5 yr
Canadian trial	cT2b–4	64 P	RT vs RT + nAD vs RT + n/c/aAD (6 mo)	22% vs 72% vs 90%	NR	24 mo
Canadian trial	—	66–67 P	RT + nAD 3 mo vs 8 mo	58% vs 65%	81% vs 79%	6.6 yr
ADJUVANT						
RTOG 85-31	cT3, pT3 or N1	44–46 WP; 65–70 P	RT vs RT + aAD (indefinite)	8% vs 32%	38% vs 53%	7.3 yr
EORTC 22863	T1–2 grade 3; T3–4	50 WP; 70 P	RT vs RT + aAD (3 yr)	40% vs 74%	62% vs 78%‡	66 mo
Swedish trial	T1–4, pN0–3, M0	50 WP; 65 P	RT vs RT + orchiectomy	39% vs 69%	38% vs 61%*	9.3 yr
RTOG 92-02	T2c–4, PSA <150	44–50 WP; 65–70 P	RT + n/cAD vs + n/c/aAD (2 yr)	28% vs 46%	79% vs 80%	5.8 yr
RTOG 94-13	PSA <100, >15% nodal risk	50.4 WP; 70.2 P	n/cAD + WP; n/cAD + PRT; WP + aAD; PRT + aAD (4 mo)	31–38%	67%, 69%, 59%§ , 68%	84 mo

*P = .02. †P = .12; differed only for Gleason 2–6. ‡P = .0002. §Lower survival with WP + aAD (P = .027; updated analysis, 2007). RTOG 86-10 data updated 2008. aAD, adjuvant AD; cAD, concurrent AD; nAD, neoadjuvant AD; P, prostate; PRT, prostate-only RT; WP, whole pelvis; NR, not reported.

- **What the table is for:** to set **short neoadjuvant/concurrent AD** (86-10, 94-13) against **long adjuvant AD** (85-31, 22863, 92-02).
- **Pattern:** DFS improves in every trial. **OS improves clearly only with long AD in high-risk cohorts** (EORTC 22863; Gleason 8–10 subsets of 85-31 and 92-02).
- **Trap:** RTOG 94-13's whole-pelvis + adjuvant arm had the **worst OS**. More field plus later AD is not better.

WHERE BODY TEXT AND TABLE 164.10 GIVE DIFFERENT VALUES

Endpoint	Body text	Table 164.10	Reason
EORTC 22863 5-yr OS	79% vs 62%	78% vs 62%	Different reports: original (1997) vs updated (2002) data cut
EORTC 22863 DFS	85% vs 48%	74% vs 40%	Different data cut (2002 update, median 66 months)
RTOG 85-31 OS	8-yr 49% vs 42% (not significant)	53% vs 38%	Different report and time point (2003 report, median 7.3 years)
3 vs 8 months NAD 5-yr DFS	71% vs 42% (8 vs 3 mo)	65% vs 58%	Text figure is probably the high-risk subgroup; the table gives the whole cohort

DURATION OF AD WITH RT: TRIALS COMPARED

Trial	Population	Comparison	Verdict
RTOG 86-10	Bulky cT2–T4	4 mo vs none	Better control and DSM; OS unchanged
TROG 96.01	cT2b–4N0	3 vs 6 mo vs none	6 mo reduced all-cause mortality ; 3 mo did not
RTOG 9910	Intermediate risk	8 vs 28 wk NAD	No difference
RTOG 92-02	cT2c–T4	4 vs 28 mo	Better except OS; OS gain in Gleason 8–10
EORTC 22863	High risk	3 yr vs none	OS gain
18 vs 36 mo	High risk	18 vs 36 mo	OS comparable ; 18 mo better QoL
High-dose RT trial	Intermediate/high	2 yr vs short concurrent	2 yr better, mainly high risk

Radiation Therapy and Chemotherapy

- **Less well studied than chemotherapy with surgery.**
- **Estramustine + vinblastine with RT (65–70 Gy):** clinical control was reasonable at 5 years, but only about half had PSA <4 ng/mL. **At 75.6 Gy in high risk** (phase II), there was only modest grade 2 toxicity, no late grade 3–4 toxicity, and 2-year BDFS of 60%.
- **Estramustine + etoposide** before and during RT: reasonable 3-year DFS and OS, with biopsy-proven local control in most at 18 months.
- **Concurrent docetaxel + RT (70.2 Gy), phase I:** maximum tolerated dose 20 mg/m². The main toxicities were **diarrhoea and dysuria**.
- **Randomised systemic intensification with RT** is summarised in the table below. Details not in the table: RTOG 0521 was over half Gleason 9–10, which may explain why it succeeded where **adjuvant docetaxel alone after RT + AD** (intermediate and high risk) did not. GETUG 12 (cT3/T4, Gleason ≥8, PSA ≥20 or pN+) added no toxicity or second cancers.
- **STAMPEDE** (about half high-risk or N+M0, mostly RT): it is unresolved whether the benefit needs aggressive treatment of the prostate, or is confined to metastatic rather than locally advanced disease. **High-risk M0 eligibility for the abiraterone comparison:** pelvic nodes, or two of **Gleason ≥8, cT3–4, PSA ≥40 ng/mL**.
- **NCCN 2023: RT + AD with 2 years of abiraterone** for very high-risk prostate cancer.

SYSTEMIC INTENSIFICATION WITH RT: TRIALS COMPARED

Trial	Added agent	Backbone	Verdict
RTOG 9902	Paclitaxel + estramustine + etoposide	Long-term AD + RT	No benefit ; closed early (thromboembolism)
RTOG 0521	Docetaxel	Long-term AD + RT	OS HR 0.69

Trial	Added agent	Backbone	Verdict
Adjuvant docetaxel trial	Docetaxel (six cycles)	RT + AD	No biochemical benefit
GETUG 12	Docetaxel + estramustine	Prolonged AD (mostly + RT)	Relapse-free 62% vs 50% (HR 0.71)
STAMPEDE	Docetaxel or abiraterone	AD (mostly + RT)	OS and failure-free survival improved
STAMPEDE (M0 high risk)	Abiraterone 2 yr (± enzalutamide)	AD 36 mo + RT	MFS HR 0.53; OS HR 0.6 → NCCN very high risk

Concept — RT for locally advanced disease is a combined-modality treatment. *The dose must be high, AD must be added and prolonged in proportion to risk, and systemic intensification (docetaxel, abiraterone) pays off only when the disease burden is high enough to have systemic reach.*

FOCAL ABLATIVE THERAPY

- **Minimally invasive, usually focal ablation** to spare whole-gland side effects; **cryotherapy and high-intensity focused ultrasound (HIFU)** are commonest; photodynamic therapy, focal laser ablation and irreversible electroporation need validation. Technologies are not rigorously compared and **high-risk data are limited**.

Cryoablation

- **Primary cryoablation (many high-risk men):** 5-year BDFS (PSA <1 ng/mL) about 40%, and about 35% with PSA ≥10 and/or Gleason ≥8.
- **With stricter thresholds (0.5 or 0.2 ng/mL), 2-year biochemical failure exceeds 80%.**
- **Randomised trial, cT2c–T3, 6 months of AD + cryoablation vs RT (66 Gy):** DSS and OS did not differ, but cryoablation gave shorter PSA-free survival (28 vs 41 months) and more biochemical failure (87% vs 53%).

High-Intensity Focused Ultrasonography

- **Data are confounded** by frequent omission of risk stratification and common use of AD to reduce prostate size.
- **AD + HIFU, locally advanced:** good 1-year control (PSA <0.3 ng/mL, negative sextant biopsies in most; no clinical progression).
- **NAD improves HIFU** in D'Amico high risk (fewer positive biopsies and biochemical failures); worse 3-year DFS independently with **stage ≥cT2c (HR 2.34) and no NAD (HR 2.19)**. As with RT, NAD helps mainly intermediate and high risk.
- **UK HIFU registry:** D'Amico high-risk men had about 65% failure-free survival at 7 years.

CRYOABLATION VERSUS HIFU IN HIGHER-RISK DISEASE

Cryoablation	HIFU
Randomised against RT: equal survival, worse biochemical control	No randomised comparison; confounded series
Outcome collapses with strict PSA thresholds	NAD independently improves DFS ; stage ≥cT2c predicts failure

Concept — Ablation behaves like RT: *local energy alone underperforms in high-risk disease and does better with added AD. Judged by thresholds as strict as those used after surgery, cryotherapy's apparent control largely disappears.*

ANDROGEN DEPRIVATION AND ITS TIMING

- **VACURG I** (placebo vs DES 5 mg vs orchiectomy ± DES): the absent OS benefit was **not explained by DES cardiovascular deaths alone**, because orchiectomy + placebo also did not beat placebo.
- **VACURG II** (DES 0.2, 1 or 5 mg): 1 and 5 mg delayed stage III progression. Immediate oestrogen was **most beneficial under 75 years with Gleason 7–10**.
- **VACURG III** (stage II): 5-year survival 48% with placebo vs 75% with DES 1 mg.
- **VACURG overall**: early AD helps progression, and possibly survival, **only in aggressive subsets; avoid oestrogen** (DES cardiotoxicity at higher doses). PSA kinetics and imaging now time delayed AD better.
- **MRC** (orchiectomy or LHRH agonist; about half locally advanced): **two-thirds of deaths were from prostate cancer**; the reduction in cancer deaths with immediate AD was **mainly in M0 disease**.
- **MRC as benchmark (cT3M0)**: median time to cancer death 90 months and OS 64 months, both not reached in EORTC 22863 → **part of the AD + RT benefit may be AD alone**.
- **SPCG-7/SFUO-3** (Scandinavian Prostate Cancer Group / Swedish Association for Urological Oncology; 78% cT3): total AD then continuous flutamide ± RT; overall mortality RR 0.68; **PSA recurrence 75% with AD alone vs 26% with RT**. Limitations: unconventional endocrine regimen, open design.
- **MRC PR07** (more advanced disease; pelvic nodes irradiated): DSS HR 0.54 at 7 years.
- **Immediate vs deferred subcapsular orchiectomy** (men unsuitable for or unwilling to have local therapy; two-thirds ≥T3; median PSA >46): most deferred men who never needed AD died of other causes. **Median time to deferred AD 3.2 years**, prompted mostly by **skeletal metastases and ureteric obstruction**; trend to better DSS with immediate AD (HR 0.63, not significant).
- **Bicalutamide 150 mg** (to spare castration effects), locally advanced: survival HR 0.68 vs placebo; progression risk down 43% overall, **HR 0.4 in locally advanced**.
- **Early Prostate Cancer programme** (three trials): better PFS, **no OS difference**; locally advanced without local therapy appeared to benefit, but **primary bicalutamide in localised disease raised death risk (HR 1.23)** vs surveillance — inappropriate at low risk, cautious use at high risk.
- **Adjuvant flutamide 750 mg after RP (pT3–T4N0)**: recurrence-free HR 0.51, **OS unchanged (HR 1.04)**; toxicity caused nearly half of withdrawals.

EARLY VERSUS DEFERRED AD, AND AD ± RT: TRIALS COMPARED

Trial	Question	Verdict
VACURG I	DES 5 mg / orchiectomy vs placebo	Less progression; no OS gain
VACURG II	DES dose	1 mg improved OS ; 5 mg cardiovascular deaths
VACURG III	DES 1 mg in stage II	Better 5-yr survival
MRC	Immediate vs delayed AD	Fewer cancer deaths (mainly M0); OS improved
Immediate vs deferred orchiectomy	Men declining local therapy	OS equal; 42% of deferred never needed AD
SPCG-7/SFUO-3	AD ± RT	RT cut cancer-specific (RR 0.44) and overall mortality
MRC PR07	AD ± RT	OS HR 0.77 ; cancer death HR 0.46
Early Prostate Cancer programme	Bicalutamide 150 mg vs placebo	PFS gain; harm in localised disease

KEY POINTS: ANDROGEN DEPRIVATION

- Early AD may improve survival in selected patients.

- **Alternative androgen manipulation** (antiandrogen, intermittent) remains investigational.

Intermittent Androgen Deprivation

- **Aim:** less AD morbidity. **Protocol:** total AD → **PSA <4 ng/mL at 24–32 weeks** → stop at 36 weeks → monitor → **restart at the pretreatment PSA (if <15 ng/mL) or at 15 ng/mL.**
- **Off-treatment intervals shorten each cycle** (about 46 → 37 → 23 weeks). Non-metastatic men: no clinical progression, PSA failure 11%, all alive.
- **Optimal timing is undetermined**, particularly in light of the trial of intermittent vs continuous AD in metastatic hormone-sensitive disease.

Quality of Life

- **Early AD must be balanced against its side effects.** Effects on bone, cognition and quality of life have only recently been appreciated.

BICALUTAMIDE 150 MG VERSUS CASTRATION

Bicalutamide 150 mg	Castration
Better sexual interest and physical capacity; more gynaecomastia, breast pain, asthenia; trend to lower overall quality of life	More hot flushes

- **Intermittent AD:** off-treatment cycles improved potency and social and family well-being.
- **AD + RT adds toxicity:** NAD → later grade ≥2 gastrointestinal and genitourinary morbidity; adjuvant AD → late grade 2–4 rectal toxicity.
- **Sexual function (a contrast):** potency similar with or without AD in RTOG 86-10 (81% vs 74%), and AD added nothing to post-RT decline elsewhere; yet after 3D-CRT **NAD independently predicted erectile dysfunction**. Morbidity may track **AD beyond 6–9 months**. Overall 5-year toxicity after 3D-CRT was <10%.
- **Number-needed-to-treat model** (cT2c–T4N0–1): **long-term AD + RT benefits outweigh side effects** and beat short-term AD on virtually every endpoint.
- **Cardiovascular:** adjuvant AD in RTOG 85-31 (median 4.2 years) did **not** raise cardiovascular mortality; short-term NAD in RTOG 86-10 did not raise fatal cardiac events.

Concept — Early AD buys progression-free time, not reliably survival. *Survival gains appear only in aggressive subsets and when AD is paired with local RT. AD alone neither cures nor controls the primary tumour, and its toxicity has to be paid for with a real benefit.*

MANAGEMENT OF DELAYED SEQUELAE

- **RP and RT are hard to compare** (mostly historical data). **Leaving the prostate in situ** matters because local control may affect survival and local recurrence may need further treatment: **10-year local recurrence after RT for cT3 is 24–74%.**
- **After RT alone: TURP (transurethral resection of the prostate) for obstruction in about a third,** hydronephrosis 20%, incontinence 13%. Untreated stage C (older series): infection 80%, **outlet obstruction 75%**, gross haematuria 45%, **ureteric obstruction 40%.**
- **AD does not prevent local progression:** about a third needed TURP 60 days after orchiectomy; orchiectomy, RT or both gave comparable local control; **with early or deferred AD over half needed TURP**, and deferred AD was prompted by ureteric obstruction (25%), anticipated local complications (10%) and rectal infiltration (4%).

- **Stage and quality of life** (Functional Assessment of Cancer Therapy [FACT]; UCLA Prostate Cancer Index urinary subscale; Hospital Anxiety and Depression Scale [HADS]): **higher stage worsens nearly every FACT subscale, mainly physical, with no link to HADS psychological symptoms.**
- **RT monotherapy, high risk:** bowel function near normative but **more bowel bother and diarrhoea;** progression or need for AD worsened sexual, urinary, social function and sleep.
- **CaPSURE:** AD with surgery or RT → sexual loss **recovering within 9 months;** **EBRT + brachytherapy → continuing decline in urinary function and bother,** worse bowel at 1 year and urinary irritation through 3 years vs EBRT alone.
- **SWOG 8794:** adjuvant RT worsened bowel and urinary function, **not erectile function;** more symptom distress; **global quality of life initially worse, then steadily improved.**
- **NCCN high/very high risk:** more intense therapy → worse urinary symptoms at 1 year, **converging** later.

Concept — A prostate left untreated or unremoved causes its own morbidity (obstruction, haematuria, ureteric obstruction) that AD does not prevent. This is the functional argument for definitive treatment of the primary, and it has to be weighed against the added toxicity of multimodal therapy.

CLINICAL TRIALS

- **Suboptimal outcomes → consider trial enrolment** (clinicaltrials.gov).
- **SPCG-15** (newly diagnosed cT3N0M0, PSA <100 ng/mL): **RP vs RT + AD, with disease-specific survival as the primary endpoint.**
- **Genomic Umbrella Neoadjuvant Study:** AD + apalutamide while tumours are sequenced, then targeted agents added (PARP inhibitor for DNA repair defects, PD-L1 inhibitor for microsatellite instability).
- **Open questions:** type and timing of combination treatment, role of chemotherapy, biomarker-directed selection.

TABLE 164.11 CURRENT AND PENDING PHASE III CLINICAL TRIALS IN LOCALLY ADVANCED/HIGH-RISK PROSTATE CANCER

Setting	Trial	Key eligibility	Question
Radical prostatectomy	AUO AP 55/09 (SEAL)	Gleason 7–10 or PSA >10	Limited vs extended pelvic lymphadenectomy
	PROTEUS (neoadjuvant)	Gleason ≥4+3 with ≥6 positive cores; Gleason ≥4+3, ≥3 cores and PSA ≥20; Gleason ≥9; or ≥2 cores of Gleason ≥8 each ≥80% involved	NAD ± apalutamide
	ECOG ERADICATE (adjuvant)	Decipher ≥0.6 and CAPRA-S ≥3	1 year AD ± darolutamide
	NRG-GU002 RADD (adjuvant)	Gleason ≥7, cN0/x, persistent PSA ≥0.2 after RP	AD + RT ± docetaxel
	AFU-GETUG 20 (adjuvant)	PSA <0.1 after RP and pT3b, Gleason >7, or Gleason 7 with tertiary 5	Surveillance vs 24 months AD
External-beam RT	RTOG 0924	Gleason 7–10, cT1c–2b, PSA <50; or Gleason 6 with cT2c–4 or >50% cores, PSA <50; or Gleason 6, cT1c–2b, PSA >20	NAD + prostate/SV RT vs whole-pelvic RT
	GETUG-AFU 18 (fully accrued)	cT3–T4, PSA >20 or Gleason 8–10; N0	AD 3 yr + 70 vs 80 Gy
	PEACE 2	Two or more of cT3–T4, Gleason ≥8, PSA ≥20	Adding pelvic RT and cabazitaxel to prostate RT + 3 yr AD

Setting	Trial	Key eligibility	Question
	ATLAS (adjuvant)	Gleason ≥ 8 and $\geq cT2c$, or Gleason ≥ 7 with PSA ≥ 20 and $\geq cT2c$	74–80 Gy + 30 mo AD $\pm$ apalutamide
	RTOG 1115 (abstract presented)	High-risk Gleason/PSA combinations, PSA ≤ 150	AD 24 mo + dose-escalated RT $\pm$ TAK-700
	ENZARAD	Gleason 8–10, or Gleason 4+3 with cT2b–4 and PSA >20 , or cN1	RT + 24 mo AD $\pm$ enzalutamide

- **Pattern:** almost every open question is about **adding a newer hormonal agent (apalutamide, darolutamide, enzalutamide, TAK-700) or chemotherapy to a standard backbone**, or about **field and dose** (lymphadenectomy extent, pelvic RT, 70 vs 80 Gy).